Lab 1:To implement Linear Regression for predicting continous target variables and evaluate the

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Pre-Lab: Understanding Google Colab and Python Libraries for Machine Learning,[NumPy, Pandas,Matplotlib,Seaborn,Sckit-learn]

```
import numpy as np
#create arrays
a = np.array([1, 2, 3, 4])
b = np.zeros((2,3))
c = np.ones((2,3))
d = np.random.rand(2,3)
print(a)
print(b)
print(c)
print(d)
```

```
[1 2 3 4]
[[0. 0. 0.]
 [0. 0. 0.]]
[[1. 1. 1.]
 [1. 1. 1.]]
[[0.65256791 0.6634882  0.5033757 ]
 [0.49200901 0.62903725 0.9802838 ]]
```

```
# array operations
print(a.shape)
print(a.mean)
print(a.std())
print(np.sum(a))
print(np.max(a))
```

```
(4,)
<built-in method mean of numpy.ndarray object at 0x7ce80ac7b4b0>
1.118033988749895
10
4
```

```
# matrix opertions
x = np.array([[1, 2], [3, 4]])
y = np.array([[5, 6], [7, 8]])
print(np.dot(x, y))
print(np.transpose(x))
```

```
[[19 22]
 [43 50]]
[[1 3]
 [2 4]]
```

```
import pandas as pd
# create DataFrame
data = {
    'Age': [25, 30, 45],
    'Salary': [30000, 40000, 50000],
    'Department':['IT', 'HR', 'Finance']
}
df = pd.DataFrame(data)
print(df.describe())
```

```
             Age   Salary
count   3.000000      3.0
mean   33.333333  40000.0
std    10.408330  10000.0
min    25.000000  30000.0
25%    27.500000  35000.0
50%    30.000000  40000.0
75%    37.500000  45000.0
max    45.000000  50000.0
```

```
# Handling missing values
df['salary'].fillna(df['salary'].mean(), inplace=True)
```

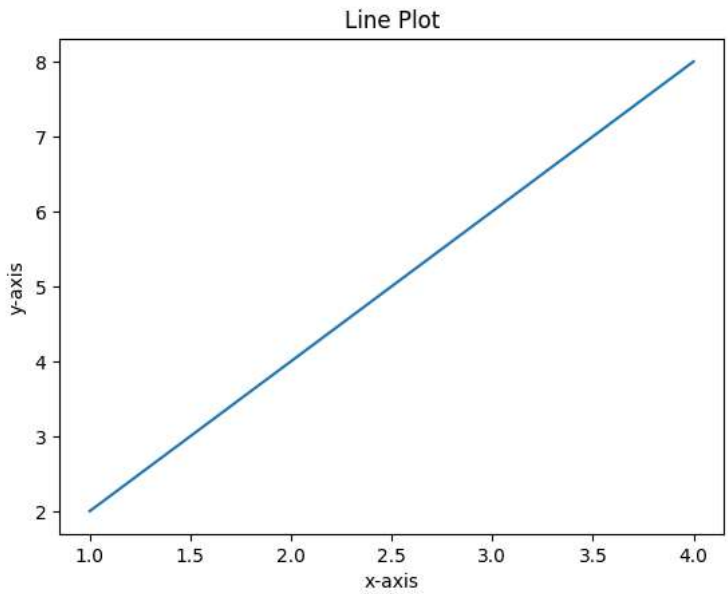
```
import pandas as pd
# encoding categorial data
data = {
    'Age': [25, 30, 45],
    'Salary': [30000, 40000, 50000],
    'Department':['IT', 'HR', 'Finance']
}
df_encoded = pd.get_dummies(df, drop_first=True)
print(df_encoded)
```

```
     sepal_length  sepal_width  petal_length  petal_width  species_versicolor  \
0             5.1          3.5           1.4          0.2               False
1             4.9          3.0           1.4          0.2               False
2             4.7          3.2           1.3          0.2               False
3             4.6          3.1           1.5          0.2               False
4             5.0          3.6           1.4          0.2               False
..            ...          ...           ...          ...                 ...
145           6.7          3.0           5.2          2.3               False
146           6.3          2.5           5.0          1.9               False
147           6.5          3.0           5.2          2.0               False
148           6.2          3.4           5.4          2.3               False
149           5.9          3.0           5.1          1.8               False

     species_virginica
0                False
1                False
2                False
3                False
4                False
..                 ...
145               True
146               True
147               True
148               True
149               True

[150 rows x 6 columns]
```

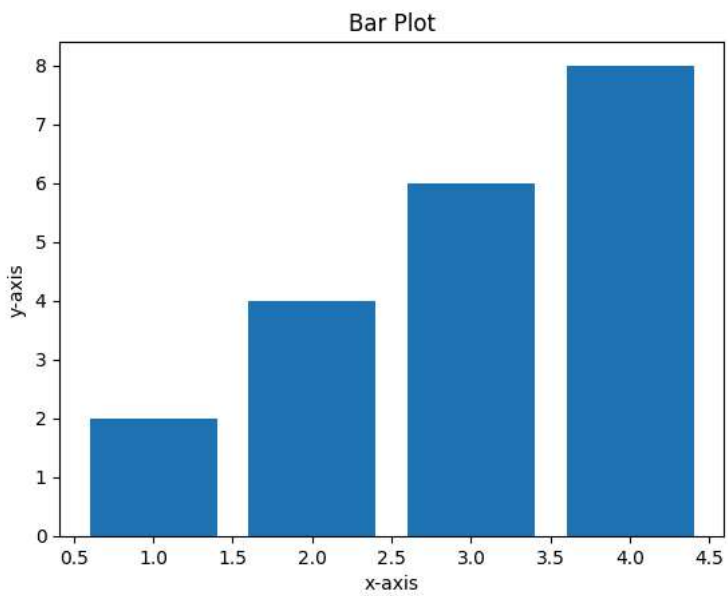
```
df_encoded = pd.get_dummies(df, drop_first=True)
print(df_encoded)
```

```
     sepal_length  sepal_width  petal_length  petal_width  species_versicolor  \
0             5.1          3.5           1.4          0.2               False
1             4.9          3.0           1.4          0.2               False
2             4.7          3.2           1.3          0.2               False
3             4.6          3.1           1.5          0.2               False
4             5.0          3.6           1.4          0.2               False
..            ...          ...           ...          ...                 ...
145           6.7          3.0           5.2          2.3               False
146           6.3          2.5           5.0          1.9               False
147           6.5          3.0           5.2          2.0               False
148           6.2          3.4           5.4          2.3               False
149           5.9          3.0           5.1          1.8               False

     species_virginica
0                False
1                False
2                False
3                False
4                False
..                 ...
145               True
146               True
147               True
148               True
149               True

[150 rows x 6 columns]
```

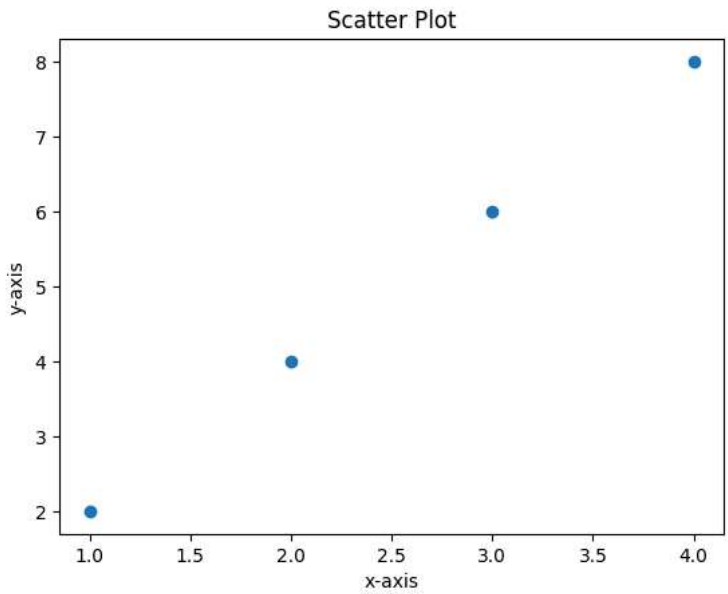
```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4]
y = [2, 4, 6, 8]
plt.plot(x, y)
plt.xlabel('x-axis')
plt.ylabel('y-axis')
plt.title('Line Plot')
plt.show()
```



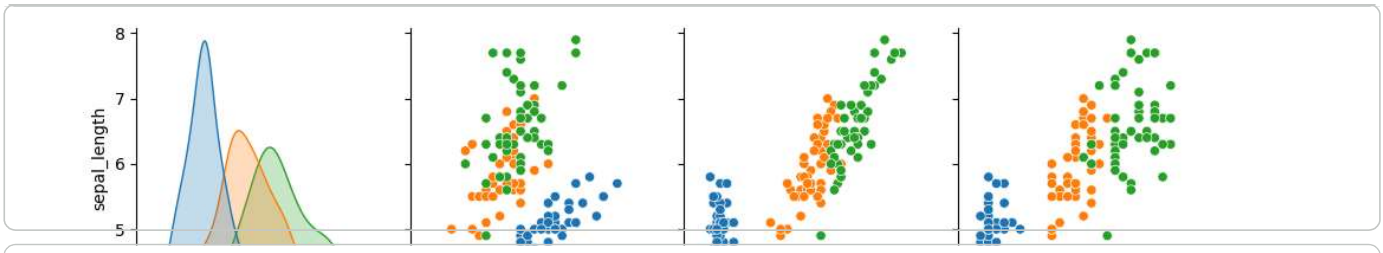

```
# bar plot
plt.bar(x, y)
plt.xlabel('x-axis')
plt.ylabel('y-axis')
plt.title('Bar Plot')
plt.show()
```



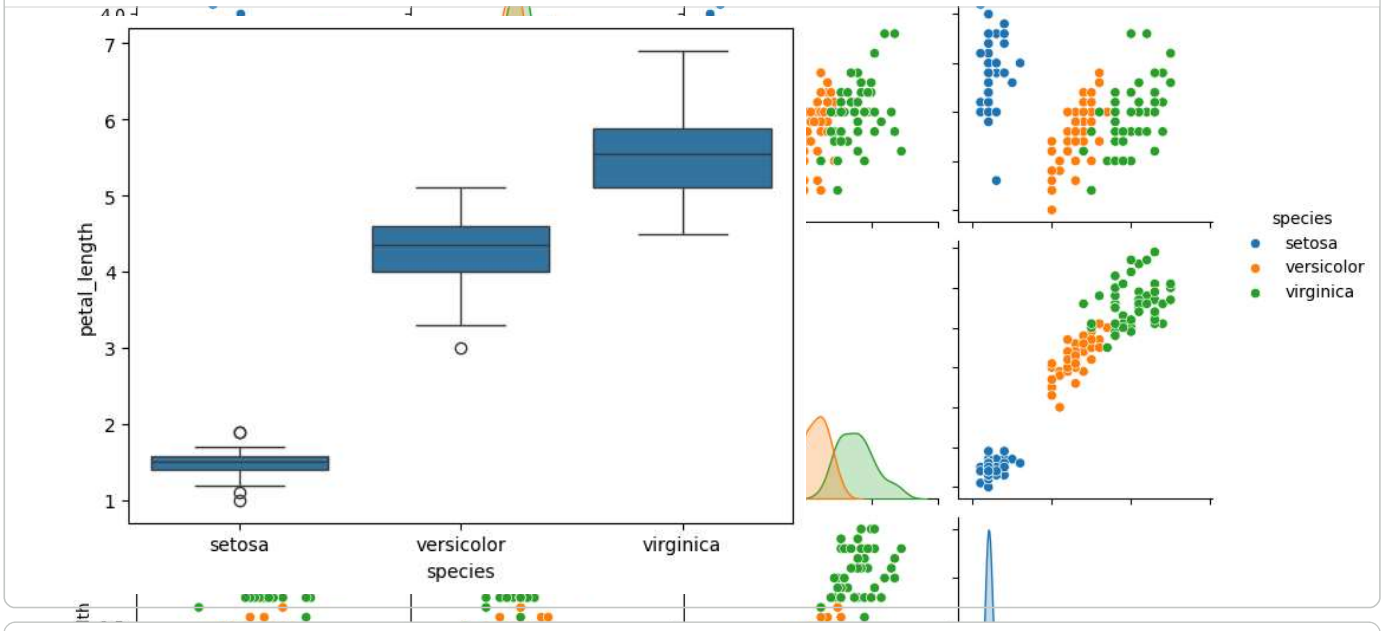

```
# scatter plot
plt.scatter(x, y)
plt.xlabel('x-axis')
plt.ylabel('y-axis')
plt.title('Scatter Plot')
plt.show()
```

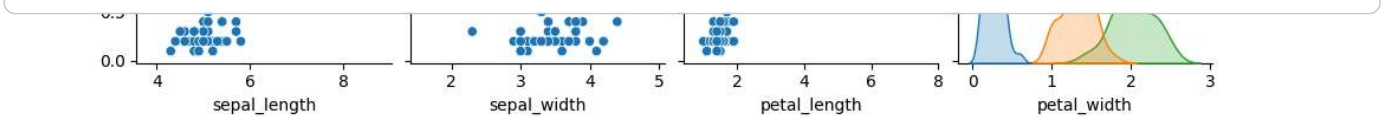



```
import seaborn as sns
import matplotlib.pyplot as plt
# load sample dataset
df = sns.load_dataset('iris')
# create pair plot
sns.pairplot(df, hue='species')
plt.show()
```




```
# Box plot
sns.boxplot(x='species', y='petal_length', data=df)
plt.show()
```




```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```